



Am stressed, must travel: The relationship between mode choice and commuting stress



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ABSTRACT

The stress of commuting has serious public health and social implications. By comparing stress across different modes it is possible to determine which modes are more heavily contributing to this potential health and social issue. This study uses a large-scale university travel survey to compare commuter stress across three modes of transportation (walking, driving, and using public transit). It also investigates the specific factors that contribute to stress using these modes. Using ordered logistic regressions, the study develops a general model of stress and three mode-specific models. Results show that driving is the most stressful mode of transportation when compared to others. We also find that stressors for some modes are not stressors for others. Knowing which specific factors make certain modes stressful will help transportation and public health professionals make commuting a safer, more enjoyable, and less stressful activity; in turn this could mitigate the potentially serious health outcomes of a stressful commute.

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1. Introduction

Many recent studies in transportation have focused on the personal experience of commuters. A person's satisfaction with their trip, overall life satisfaction, and stress experienced while commuting have become increasingly crucial parts of our understanding of travel behavior, especially regarding mode-choice. Mode-switching may be inhibited because certain physical and mental health implications are associated with a mode (Abou-Zeid, Witter, Bierlaire, Kaufmann, & Ben-Akiva, 2012). Findings from studies that focus on the personal experiences of commuters can have real policy implications by suggesting that factors that may inhibit the uptake of more sustainable modes of transportation should be minimized, while factors that make these modes more attractive to users should be explored.

Stress is one of the most serious physical and mental health implications of commuting. Almost all commuting can be stressful—rushing to get to work or school in the morning is often an unpleasant experience—and some modes may be causing more stress than others. Discovering the mode-specific factors that contribute to a stressful commute highlights where policy focused on increasing sustainable mode-share can be effective. Sustainable mode use can be made more attractive by minimizing the factors that make sustainable modes stressful. Perhaps more importantly, commuting is almost ubiquitous: a hefty share of any population travels daily and, correspondingly, the stress experienced while commuting affects a large number of people.

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Factors contributing to stress during a commute can be broadly grouped into two categories (Novaco, Stokols, & Milanese, 1990). First, there are objective or environmental stressors. These stressors negatively impact a person's control or comfort while commuting. Second, there is the subjective experience of these stressors, which are influenced by (for example) the satisfaction a person has with a mode. These personal subjective factors act as a filter through which objective stressors are experienced. This interaction between personal experience and objective stressors is derived from the work of Novaco et al. (1990) and Koslowsky (1997), and is shown in Fig. 1.

In order to better understand the factors leading to a stressful commute, this study uses a large-scale university travel survey to compare commuter stress across three modes of transportation (walking, driving, and using public transit) during a cold snowy day. By highlighting which factors lead to stress on different modes, transportation planners, engineers and policy makers can better understand the factors that can make the commute more enjoyable and provide a less stressful experience.

2. Literature review

What is stress? Lazarus and Launier (1978) define stress as a situation where the environment overwhelms the person: “these relationships refer neither to person nor environment as separate variables, but they describe a balance of forces such that environmental demands tax or exceed the resources of the person.” Other seminal work on stress echo this definition (see, for instance, Fink, 2000; Selye, 1976). Furthermore, commuting has been linked to stress numerous times (for a review see, Novaco & Gonzales, 2009). The experience of being stuck in traffic or waiting for a delayed train is understandably a hardship and quite common. These and other demands experienced while commuting often lead to stress. Yet, the relationship between commuting and stress is not clear. For example, just because a delay is experienced does not mean that the user experiencing that delay is necessarily stressed.

2.1. The effects of stress

Commuting stress (especially if it is unduly associated with specific modes) may inhibit people switching to a more sustainable mode of transportation. Certainly, this is a concern. Also of concern are the potential public health effects of stress, and, consequently, the serious health and life satisfaction implications of stressful commuting. Longer commutes by car, for instance, have been related to an increased risk of heart attack and obesity (Hoehner, Barlow, Allen, & Shootman, 2012). In particular, the stress of commuting has been linked to poor quality sleep, exhaustion, depression, and feelings of poor health (Gee & Takeuchi, 2004; Hansson, Mattisson, Björk, Östergren, & Jakobsson, 2011). Commuting stress has also been shown to negatively impact a person's ability to focus or complete tasks (Wener, Evans, & Boatley, 2005). Poor job performance and shortened job tenure, no doubt exacerbated by these health and mental effects, are also linked to commuting stress (Koslowsky, Kluger, & Reich, 1995; Novaco et al., 1990).

Nevertheless, the view that commuting, and travel in general, is an unavoidable burden has come under much criticism. Mokhtarian and Salomon (2001) have argued that travel is not always a derived demand. Rather, travel, including commuting, can be enjoyed for its own sake. A recent study showed that travel is linked to higher life satisfaction (Ory et al., 2004). Morris and Guerra (2014) have argued that those who are happy are more likely to travel. A recent Swedish study demonstrated that feelings while commuting are generally positive or neutral (Olsson, Garling, Ettema, Friman, & Fujii, 2013). Enjoying one's commute may be linked to notions of a desired minimum distance between a person's home and work locations. Commuting time may be enjoyed as a time to decompress or unwind, and as a necessary transition between home and work. Interestingly, this may have an effect on stress as well. Those who have no commute report being more stressed than

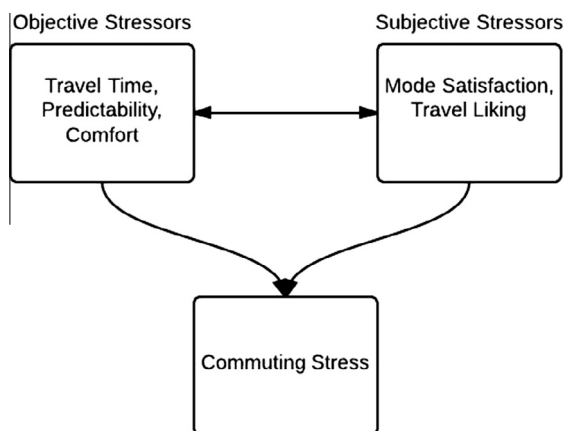


Fig. 1. Study framework.

those who commute less than thirty minutes (Haider, Kerr, & Badami, 2013). In this way, commuting, as an activity that both positively and negatively affects the commuter, is a significant player in a person's overall subjective well-being (Reardon & Abdallah, 2013).

2.2. The causes of stress

There are two major objective variables that help explain the forces that cause stress: measurements of control and comfort. These two variables come out of the wealth of literature focusing on the impedance model of commuting stress, first developed by Novaco, Stokols, Campbell, and Stokols (1979). Travel impedance is the notion that specific stressors (traffic congestion, for instance) *impede* one's commute. At its most basic form, the degree of impedance is defined by the distance traveled and the time it took to travel that distance; high impedance occurs when one is traveling slowly (Novaco et al., 1979). Recent studies have also investigated how lengthy travel distances relate to commuting stress (Ettema, Garling, Olsson, & Friman, 2010; Stutzer & Frey, 2008). However, the stress of this impedance is mediated by a variety of factors. Control is seen as a mediator of commuting stress. The level of impedance during the trip may be a cause of stress, but having personal control of one's own trip lessens this stress (Novaco et al., 1990; Schaeffer, Street, Singer, & Baum, 1988).

Yet, the task of determining a person's level of control while travelling is not straightforward. A number of factors have been used as proxies for personal control, such as travel predictability (Evans & Wener, 2002). A commute is seen to be less predictable when factors outside of a person's control (traffic congestion, train delays) occur (Wener, Evans, Phillips, & Nadler, 2003). For public transit, transfers are also seen as a cause for less predictability, and thus less control (Wener et al., 2005). However, Wener et al. (2005) also argue that shorter travel times mitigate the stressful effects of transfers. One recent study used self-reported ratings of predictability to determine this variable, using questions concerning consistency and predictability of arrival time (Sposato, Röderer, & Cervinka, 2012). In addition to predictability, having a variety of commuting options, such as different driving routes, have been posited as mitigating the stress of impeded travel (Novaco et al., 1990).

Comfort, or the lack of it, is also seen as a mediator of impedance. Discomfort from crowding has long been associated with commuting stress, especially the stress associated with public transportation (Koslowsky et al., 1995; Lundberg, 1976). Uncomfortable heat or noise is also seen to cause stress while travelling (Novaco & Gonzales, 2009; Wyon, Wyon, & Norin, 1996).

These physical or environmental occurrences, which result in a slower moving trip, lack of control, or discomfort, are objective stressors that lead to a stressful commute. The subjective experience of these stressors, however, is a crucial filter between the occurrence and the stress experienced (refer to Fig. 1, above). Ory et al. (2004) argues that there is a difference between objective mobility (the actual distance and frequency of travel using a certain mode) and subjective mobility. Both objective measures of mobility and self-reported subjective measures of mobility (responses to questions like "how often do you feel like you travel") have an effect on a person's "travel-liking" (Ory et al., 2004).

Novaco et al. (1979, 1990); and Koslowsky (1997) argue that the stress of commuting is similarly derived not just from objective impedance but also from the mediating experience of subjective impedance. How one feels about one's trip has a mediating effect on one's experience of impedance, and thus an effect on stress. Subjective ratings by the commuter regarding their satisfaction with the commute, the pleasantness of the trip (Novaco et al., 1990), or feelings about bus conditions or commuting environment (Koslowsky, 1997), have all been used as measures of subjective impedance. Although both authors put forth this notion of subjective impedance, they mainly focus on the subjective experience of one or two modes, driving or taking public transit. Furthermore, their goal is not comparative; they are not focusing on the experience of driving or public transit *per se*, but are using these modes as a petri dish from which to derive a broad (mode-independent) understanding of commuting stress. This study, on the other hand, argues that mode is intimately related to commuting stress. A comparison of stress between modes and a comparison of mode-specific stressors offer insight into the causes of commuting stress.

2.3. Mode and stress

Surprisingly little work has been done comparing stress across all major modes of transportation. We argue that a broad understanding of commuting stress cannot be derived from a study of just one or two modes. Rather, mode must be understood as an important component of commuting stress: different modes have inherently different variables that contribute to the stress of that commute. Notions of control or lack of control vary whilst driving, walking, or taking transit, and discussions of comfort and crowding vary depending on which mode is under scrutiny.

Of the studies that have been done, many focus on the stressful effects of using an automobile for commuting (for instance, Novaco et al., 1979, 1990; Rasmussen, Knapp, & Garner, 2000). Some studies have compared the stress of driving to the stress of commuting on a train, finding that driving is more stressful (Wener & Evans, 2011). Very little work has been done on the stress of active modes of transportation. This relative dearth is perhaps because recent studies have linked active modes with higher overall life satisfaction (St-Louis, Manaugh, Lierop, & El-Geneidy, 2014). In one broad comparative study, Abou-Zeid (2009) found that (using self-reported stress levels on a 5-point Likert scale) active commuters were the least likely to report being stressed during their commute. Drivers, on the other hand, were the most likely to report stress. However, discovering mode-specific stressors may further our understanding of commuting stress. It is clear that commuting using a car is a radically different experience compared to walking, and, thus, the stress of these commutes occurs for different reasons.

One recent study compared stress across active, transit, and driving modes, and found that active modes of transportation are less stressful when compared to public transit or driving, and that public transit is the most stressful (Gatersleben & Uzzell, 2007). However, this study was not devoted to stress in itself, but was interested in 'affective appraisals' of a person's commute, both positive and negative. They included stress as one factor for understanding the pleasantness of a commute, and, thus, the attractiveness of the mode. They found that transit was the most stressful and least pleasant mode. Our study compares stress across modes and delves deeper into understanding the variables, both subjective and objective, that are important stressors during the use of specific modes. In this way the *reasons* behind the stress experienced on a mode can be discovered. This comprehension is crucial for understanding commuting stress and the steps that can be taken to lessen it. Furthermore, Gatersleben and Uzzell (2007) were limited by a small sample size ($n = 389$), which only consisted of university employees commuting in a semi-rural environment. Our study, focused on a large and diverse sample of commuters in an urban environment, is better suited to discovering findings pertinent for urban policy decisions.

3. Methodology

3.1. Survey

The data used for this study is compiled from a large-scale commuter survey administered at McGill University in Montreal, Canada. The survey targeted a sample of students (one-third of the student population), and all faculty and staff. Administered online, every participant was emailed an invitation with a list of prizes that were offered as an incentive. The survey was online for thirty-five days between March and April of 2013. A total of 20,851 survey invitations were distributed, with a response rate of 31.7%. This response rate is comparable to a previous study (Whalen, Paez, & Carrasco, 2013) where they had a 22% response rate. After removing partial or inadmissible cases, 5599 surveys were kept as usable responses. Because of further data requirements, discussed below, a smaller set of responses is used for this study. The survey asked for a description of respondents' commute on a typical cold snowy day and a typical warm dry day. Respondents were asked for in-depth details about each leg of their trip, specifying the mode used and the time spent using that mode. General questions regarding their satisfaction, habits, and preferences for travel were also asked, and socio-demographic information was gathered.

3.2. Sample

The final sample used for this study included 3794 responses. This sample is composed of 30% staff, 20% faculty, and 50% students. The sample includes respondents who commute to McGill's downtown campus (93.3%) and its Macdonald campus (located in a suburban area on the island of Montreal). For this study we focused our attention on 'cold-snowy day' responses. The survey was administered at the end of a long winter (snowy weather regularly continues into early April in Montreal). At the time the survey was conducted the temperature was below freezing for the majority of the time. Therefore, the responses to questions pertaining to winter conditions, especially regarding reported levels of trip satisfaction and stress, are, we feel, better representative of the levels of stress associated with commuting than questions regarding summer (warm-dry) conditions. We expect that levels of satisfaction or stress reported by respondents for warm-dry days is biased due to the fact that the survey was conducted at the tail-end of a long winter. The temperature was below freezing in Montreal between early November 2012 until March 2013 and the survey was conducted in March and April. Also, respondents who reported using more than one mode of transportation or stopping on their way to campus are excluded from the study.

Respondents are assigned a variable based on their mode of transportation. If a respondent only walked to their destination, they are deemed a pedestrian. Similarly, if they drove to McGill they are deemed a driver (in our final sample, only 11 respondents are passengers in automobiles, and are excluded from the study). Those who used public transit for one leg of their trip are deemed public transit users. We originally included a sample of bicycling respondents ($n = 64$), but this sample is excluded for being too small. Table 1 shows summary statistics regarding each of the included travel mode groups. This breakdown by mode is desired for two reasons. First, we wish to isolate the mode specific factors that contribute to commuting stress; this categorization allows us to investigate what factors affect pedestrians and compare these factors to those for drivers, for example. Second, this also allows us to compare the explanatory power of a general stress model (where all respondents are included) to mode specific models (where one model is generated for each mode).

The summary statistics shown above only include variables that are applicable across all groups (except percentage with a driver's license for drivers, which is redundant). Transit users are the most prominent group in our sample, with over a fifty percent share. Pedestrians have a higher desire to live close to McGill than the other groups (with a mean of 4.46 out of five), especially when compared to drivers. A quick comparison of the mean levels of stress across the different mode groups show that drivers report the highest mean stress, with transit users following closely behind. Drivers also tend to be older (mean of 44 years) have the highest life satisfaction of all mode groups, and tend to have a higher income than the rest of the sample, all factors that are related to less stress in our models. Interestingly, they also have the highest mean additional time budgeted, indicating that circumstances beyond a drivers' control necessitate additional time. They also seem to have lower satisfaction with their commute, indicating that subjective stressors are important components for this mode. They also tend to agree, more than the other groups, that arriving at their destination is the only important component of their trip, and they would like to use other modes more.

Table 1
Summary statistics.

N	General model 3783		Walking 1083		Driving 626		Transit 2074	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Stress	2.92	1.28	2.67	1.31	3.19	1.23	2.96	1.26
<i>Control variables</i>								
Age	35.9	13.9	29.39	11.39	44.34	13.36	36.74	13.64
Male (%)	0.39	0.49	0.39	0.49	0.42	0.49	0.37	0.48
With a Driver's License (%)	0.84	0.37	0.78	0.42	–	–	0.83	0.37
Life satisfaction (1–10)	7.36	1.7	7.38	1.66	7.62	1.59	7.28	1.74
Income (1–10)	1.93	2.33	1.12	1.92	3.68	2.9	1.83	2.05
Days per month spent on campus	0.75	0.43	0.7	0.46	0.7	0.46	0.79	0.4
"Importance of Being Close to McGill" (1–5)	3.58	1.39	4.46	0.97	3.15	1.43	3.25	1.36
<i>Objective stressors</i>								
How many modes tried in the past year	1.34	0.66	1.4	0.71	1.5	0.82	1.26	0.55
Mean commute time (minutes)	32.91	20.96	17.5	11.58	32.54	20.08	41.05	20.48
Mean additional time budgeted (minutes)	14.76	14.46	7.32	7.29	21.15	17.44	16.65	14.69
<i>Subjective stressors</i>								
"The only good thing about traveling is arriving at my destination" (1–5)	2.85	1.2	2.54	1.16	3.15	1.2	2.92	1.18
"Would Like to Walk More" (1–5)	3.24	1.21	2.94	1.16	3.72	1.17	3.25	1.19
"Would Like to use Transit More" (1–5)	2.13	1.1	1.94	1.03	2.67	1.34	2.07	1
"Would Like to Drive More" (1–5)	1.91	1.15	1.79	1.13	2.15	0.98	1.89	1.2
<i>Mode</i>								
Percentage walking	0.29	0.45						
Percentage transit	0.54	0.5						
Percentage driving	0.17	0.37						

3.3. Modeling technique

This study uses a series of ordered logit regressions to examine the relationship between subjective and objective factors, modal influence, and stress. For all models, reported stress is the dependent variable. The survey asked respondents if they agreed or disagreed (on a 5-point Likert scale) that they "felt stressed during their commute to McGill." Since this response is inherently ordinal, an ordered logit modeling technique is appropriate. For independent variables, we use pertinent variables based on previous studies concerning stress and subjective well-being. Control variables are included to accommodate for demographic differences and residential location choices. Other variables are included to capture both objective stressors and subjective experiences related to these stressors.

Basic personal data including age, gender, and income (divided into four levels) are included. Gender has been consistently included in studies of stress and stressful commuting. The common consensus is that, *ceteris paribus*, women are more likely to be stressed than men (Novaco & Collier, 1994). Age squared is tested to see if there is a change in effect among different age groups. Also, the respondent's main campus (downtown or at Macdonald) is tested to control for any effect. Similarly, the respondent's status (student, faculty, or staff) is considered. To control for the effect of residential choice (self-selection), various residential-location choice variables are tested: Importance of living close to transit, importance of living in a location where driving is not necessary, and importance of living close to campus (all rated on a 5-point Likert scale) are evaluated. Car ownership, is another variable which should be controlled for. This can be done through a variable indicating how many vehicles each household has. Also, a variable indicating how many days per month spent on campus (e.g., on campus twenty days a month or approximately five days a week) is used to see if the repetition of the commute has any effect, both positive and negative. Finally, overall life satisfaction is also included.

Two major objective impedance variables are included, the time of the commute, and, for transit users, if they transferred or not during their commute. Beyond these objective measures, a number of mediating stressor variables, approximating the level of control the person has and the level of discomfort the person is experiencing, are included. The survey asked respondents to report how much additional time they budgeted (in minutes) for their daily commute. This variable can be used to approximate how unpredictable a commute is, and, thus, how much control a person has during her trip. Another variable is included to approximate how much choice a commuter has: The survey asked respondents to count how many modes (different from their primary mode) they used in the last year to commute to school or work. A mode used at least once in the last year indicates that this mode is at least a potential or viable option for the user; those with more options available have potentially more choice and control over their commute.

Stressors dealing with the level of comfort experienced during the commute are mode-specific and hard to gather objectively. Self-rated feelings of comfort, 'safety from crime,' and 'safety from traffic' are included to gain insight into these stressors. Connected to feelings of comfort is the fact that a commute that is long or has an additional time budget that is excessive does not directly correspond to a stressful commute. As has been discussed, the subjective experience of these objective stressors is also important. To capture these subjective stressors' effects, a series of variables dealing with self-

reported feelings toward travel mode and satisfaction with specific components of that mode are included. Respondents are asked to rate their satisfaction with their commute time, cost of their commute, and other specific satisfaction variables (to be discussed below). Respondents are also asked to rate whether or not they felt that travel had a purpose besides arriving, and if they would like to use each mode more than they currently do. These variables indicate if they have any enjoyment for their trip (beyond arriving) and approximate their satisfaction with their commute and mode of the moment. These variables attempt to capture similar sentiments as the 'Travel Liking' variable used by Ory et al. (2004).

In order to come to a better understanding of the effect mode has on commuter stress, and to discover in greater detail the stressors of specific modes, a series of models is developed. First, a general model for the entire sample is generated and includes the universal (not mode-specific) variables, described above. It also includes dummy variables indicating which mode is used (walking, driving, or taking transit) to measure the effect different modes have on stress. Three further models are also developed, one for each mode under study. All the variables included in the original general model are also present in the mode-specific models, unless they are found to be insignificant. If a variable is insignificant, it is removed, and the change in the Log-Likelihood of the model as well as the change in effect of other variables is studied. If little change occurs, the insignificant variable is not included in our final analysis. In addition to these general variables, other, mode specific variables are included.

All variables originally recorded using 5-point Likert responses are recoded into dummy variables, where responses (1) and (2) are categorized as a 'Low' response, (3) as a 'Medium' response and (4) and (5) as a 'High' response.

3.3.1. Walking model

In the walking model, only those respondents who used walking as their primary mode of commuting to campus are included. For walking specific variables, 'safety from traffic' and satisfaction with comfort are included. Safety from crime, having a driver's license, days spent on campus, and the number of viable options available are also tested.

3.3.2. Driving model

Similar to the walking model, only those respondents who used an automobile as their primary commuting mode are included, and these respondents are only drivers. A very small sample size of automobile passengers present in the sample ($n = 11$) limits our ability to investigate this subset of automobile users, and they are excluded. Satisfaction with time and cost are included as driving specific variables. Safety from crime, satisfaction with comfort, and satisfaction with consistency are included as well.

3.3.3. Transit model

We determine that those who use transit are those who use a public bus, train, or subway for at least on leg of their trip. Included are dummy variables for the transit type used: subway, bus, or commuter train. In addition, a dummy variable indicating if the respondent walked to the transit station (versus drove) and if the respondent used two different modes (necessitating a transfer) versus one, are included.

Subjective transit variables offer a unique challenge because the pertinent questions in the survey are transit-type specific. These questions (dealing with respondents satisfaction with time, for instance) are asked not regarding a commuter's satisfaction with time on transit in general, but satisfaction with her time on the bus, her time on the subway, or her time on the train, depending on which type(s) she used. To explore this level of detail but to avoid developing models for each transit type, we collapsed these three subsets of the transit sample into one. Most respondents did not use every transit type, and therefore did not report information on their subjective experience of unused modes. For these cases, their answers to these un-asked questions are coded as a 0. This code is categorized under the 'Low' level of dummy variables for ranked responses. In the transit model the lowest level is excluded from testing. In this manner, we have three layers of transit subjective experience, one for train, one for bus, and another for subway riders. The dummy levels that remained (Medium and High) explain the effect of *having* such an opinion for those who used that mode. These ranked variables are similar to the ones included in the pedestrian and driving models. 'Satisfaction with the consistency' of the transit type, a rating of how comfortable the type is, how satisfied the rider is with the time it takes to reach the station, and the waiting time at the station are all included. Satisfaction with time and safety from crime are also tested.

4. Results

Table 2 shows the results of our regression analyses. Prior to discussing the model results, we provide a comparison of the general model and the mode specific models. The reader should note that the general model would usually tend toward a parsimonious structure, while the mode-specific models would provide additional flexibility with higher number of model parameters. Hence, any comparisons of these model structures need to weigh the advantage of the mode specific models while penalizing for additional parameters. We employ the Likelihood Ratio (LR) test to compare the models. The LR test statistic is defined as $2 * (LL_U - LL_R)$ where LL_U and LL_R represent log-likelihood (LL) values at convergence of the unrestricted and restricted models respectively. In our case, the sum of log-likelihood values of all mode-specific models corresponds to the unrestricted model LL, while the general model LL corresponds to the restricted model LL. The LR test statistic thus computed is compared with the chi-square distribution value of k degrees of freedom where k corresponds to the additional

Table 2
Ordered regression results.

Log-likelihood		General –5419.93 B	Pedestrian –1512.59 B	Driver –855.72 B	Transit –2880.681 B
	Dummy level				
<i>Controlling variables</i>					
Age * 10 ^{–2}		–1.347***	–2.303**	–1.264	–0.573
Male		–0.512***	–0.408***	–0.406*	–0.487***
With a driver's license		–0.373***	–	na	–0.438***
Life satisfaction (1–10)		–0.157***	–0.214***	–0.055	–0.107***
<i>Income</i>					
(Comparison variable) \$0–\$19,999					
\$20,000–\$79,999		–0.304***	–0.378**	–0.083	–0.335**
\$80,000–\$119,999		–0.43***	–0.666**	–0.034	–0.565**
\$120,000+		–0.451**	–0.402	–0.31	–0.607**
Days per month spent on campus * 10 ^{–2}		–	–	–3.071**	–
Importance of being close to McGill	High	0.327***	0.643	0.316	0.292**
<i>Objective stressors</i>					
How many modes tried in the past year		–0.127**	–	–	–0.224**
Commute time * 10 ^{–2}		1.356***	1.228**	1.363**	1.031***
Additional time budgeted * 10 ^{–2}		2.69***	3.291***	2.075***	2.229***
<i>Subjective stressors</i>					
The only good thing about traveling is arriving at my destination	Low	–	–0.393**	–	–
The only good thing about traveling is arriving at my destination	High	0.494***	0.028	0.27	0.534***
Would like to walk more	Med	–0.215**	–	–	–
Would like to walk more	High	–	–	0.433**	–
Would like to use transit more	Low	–0.273***	–	–	–
Would like to use transit more	High	–	–	–	0.58**
Would like to drive more	Low	–0.285***	–	–	–
Would like to drive more	High	–	–	–	0.291*
<i>Mode</i>					
(Comparison variable) driver		na	na	na	na
Pedestrian		–0.702***	na	na	na
Transit		–0.628***	na	na	na
<i>Pedestrian and driver variables</i>					
Satisfied with time	Low	na	–	1.052***	na
Comfortable	High	na	–0.81***	–	na
Safe from traffic	Low	na	0.42***	–	na
Safe from traffic	Med	na	0.52***	–	na
Satisfied with cost	High	na	–	–0.519**	na
<i>Transit variables</i>					
Mode used subway		na	na	na	–0.282
Mode used train		na	na	na	0.068
Walked to transit station		na	na	na	–0.322*
Two modes used		na	na	na	1.188***
<i>Train satisfaction</i>					
Satisfied with the time it takes to reach the station	Med	na	na	na	–1.152*
Satisfied with the time it takes to reach the station	High	na	na	na	–0.901*
Satisfied with waiting time in station	High	na	na	na	–0.49*
<i>Subway satisfaction</i>					
Consistent	High	na	na	na	–0.337**
Comfortable	Med	na	na	na	–0.457**
Comfortable	High	na	na	na	–0.51***
Satisfied with waiting time in station	Med	na	na	na	–0.439**
<i>Bus satisfaction</i>					
Consistent	Med	na	na	na	–0.392*
Consistent	High	na	na	na	–0.295*
Satisfied with waiting time in station	Med	na	na	na	–0.44**
Satisfied with waiting time in station	High	na	na	na	–0.793***

– “found insignificant, and removed”, na “not applicable”.

* $p < 0.05$.

** $p < 0.01$.

*** $p < 0.001$.

parameters in the unrestricted model, i.e. k = sum of the number of parameters in all mode-specific models minus the number of parameters in the general model. The LL value sum for the mode specific models is -5248.99 , and the general model is -5419.93 . These values yield an LR test statistic of 341.89, which is considerably higher than the corresponding chi-square value with 26 degrees of freedom at any level of significance. A comparison of the two models using more stringent criterion such as Akaike Information and Bayesian Information criterion arrive at the same conclusion. The statistical comparison clearly highlights that in modeling commute stress, considering stress by mode offers an improved data fit.

Some of the variables originally tested are removed from the final models because of insignificance. Variables are removed if (a) they are insignificant in the model, (b) their removal does not affect the log-likelihood of the model in question, and (c) other variable coefficients are unaffected. Interestingly, most of the variables regarding residential choice (“importance of living close to transit” and “importance of being in a location where driving is not necessary”) are found to be insignificant, and removed. However, “importance of being close to campus” is significant and is discussed below. Car ownership variable “number of cars in a household” was also dropped for lack of significance. It should also be noted that one’s campus (Macdonald or downtown) or one’s status (faculty, staff, or student) are both found to be insignificant.

Most personal controlling factors have a negative effect on stress. Age is negatively associated for all models except for the transit model, where it is found not to be significant. Being male or having a higher life satisfaction are also related to less stress. Importance of residential proximity to campus, interestingly, has a significant but positive effect on stress. As a control, it is indicating that those who find it preferable to be close to campus are likely to be more stressed due to their commute while those individuals that do not consider their proximity to campus to be as important are less likely to be stressed. This finding is explained by Petter Næss’s claim that “travel attitudes are not the most important criteria of residential preferences, and several constraints can prevent people from realizing what they would otherwise prefer” (Naess, 2014, p. 70). In this study, those who indicate a desire to live closer to campus may also be indicating that, at present, they cannot. Thus, their commute is more stressful. Also, by testing various residential choice variables (two of which were found to be insignificant) we are confident that self-selection has been controlled for. Indeed, Næss (2014) finds that various residential choice variables, if included, account for little variation (around 6%), in regressions modeling residential proximity to downtown.

“Days per month spent on campus” has an influence on only those who drove. Its negative association with stress may be the result of drivers becoming more accustomed to their drive. A regularity with their commute allows them to develop coping strategies and become more adept at the drive, thus potentially mitigating the stressful effects of their trip (Naess, 2014, p. 70). Income, at all levels, has a strong negative effect on stress both in the general model, and for transit users. Interestingly, the effect is not present for drivers and less present in the pedestrian model.

To determine the general effect that mode has on stress, we include dummy variables indicating which mode is used in the general model. Both walking and taking transit have a strong negative effect on stress when compared to driving. This finding is not unanimously supported in other papers. A recent study looking at 33 urban areas in Canada found transit to be the most stressful mode (Haider et al., 2013). Ory et al. (2004) also found that transit is the most stressful. Gatersleben and Uzzell (2007) find that both driving and taking public transit are stressful when compared to active modes. These discrepancies with our study can be explained because of differences in study area. Our study is focused on a primarily urban population, with corresponding access to a relatively wide variety of public transit options (indeed, over half of our sample uses transit on a daily basis). These two other studies (Gatersleben & Uzzell, 2007; Haider et al., 2013) are focused on either a wide range of urban and semi-urban areas of Canada (many of which have limited public transportation access), or a semi-rural area of England, not a dense urban center.

Our objective stressors are aimed at measuring the direct impedance of the commute and the mediating effect control and comfort have on that impact. First, our measure of commuting options did have a negative effect in the general model. The result indicates that individuals who have used multiple mode options are likely to feel less stressed. However, in the mode specific models, this variable is only significant for transit users. This indicates that for transit users, having other (possibly non-transit) commuting options lessens the stress experienced. Commute time is found to be positively associated with stress both in the general model and across modes. Those with a lengthier commute are, on the whole, more stressed. What is interesting is that our main mediating variable, additional time budgeted, has an almost twice as large impact on stress than simple commute time in all models. The more time a respondent must budget to deal with unpredictable situations on her commute, the more stressed the respondent will be.

The subjective stressors accommodate for the impact of individual perceptions and attitudes on commute stress. Similar to control and comfort, these variables have a mediating effect between the actual objective instigator of stress (traffic congestion, for instance) and the stress actually experienced by the commuter.

Agreeing with the statement that ‘the only good thing about traveling is arriving’ is positively associated with stress in all models except for the pedestrian model, where *not* agreeing is negatively associated with stress. This result indicates that those who do not enjoy their travel for reasons other than arriving (reading, listening to music, the experience of speed, for example, are more stressed, a finding corroborated by Gatersleben and Uzzell (2007). Related to this effect are the variables concerning a respondent’s desire to use other modes more often. In the general model, not wanting to use transit or drive more is negatively associated with stress. This may indicate that if a commuter is satisfied with their mode (and therefore does not wish to use another mode more often) they are less stressed. Also, in the transit model, wanting to use transit or drive more is positively associated with stress. This indicates that, for transit commuters, not being content with one’s commuting habits (and thus wanting to use transit or drive more) leads to more stress. Similarly, for drivers, wanting to walk more than one currently does means they are more likely to be more stressed during their commute. Perhaps a component of

the stress certain drivers feel is related to feelings of control; they feel more stressed because they wish to walk to work or school, but cannot.

4.1. Mode specific variables

Comfort is a mode-specific measure: comfort while walking has nothing to do with seat availability, for instance. Also, comfort is inherently a subjective occurrence. One cannot directly measure the comfort a person is experiencing. Having a (self-reported) high comfort rating has a negative effect on stress and is significant in the pedestrian and transit models.

4.1.1. Pedestrian and driver subjective stressors

For pedestrians, satisfaction with time is not significantly associated with stress. However, not feeling safe from traffic is positively related, which indicates that perhaps the most stressful possibility of the pedestrian experience is dealing with automobiles. Our finding that time is insignificant for pedestrians corroborates previous studies that find that a pedestrian's satisfaction with their commute is related to the motivations (environment, finances, or convenience) behind their mode choice, and has less to do with the time spent walking (Mokhtarian & Salomon, 2001).

Conversely, for drivers, time satisfaction is important. Having a low satisfaction with time is strongly and positively associated with stress. Considering they have the highest mean time budget (see Table 1), time and predictability are important, and more stressful, variables for drivers when compared to other modes.

4.1.2. Transit specific stressors

For transit users, no one specific transit type (subway, train, or bus) is associated with a stressful commute. However, walking to your transit station, compared to driving, is negatively associated with stress, and having used two modes (versus one) is strongly positively associated with stress. This indicates that the unpredictability of transit commuting (which occurs when a transfer between modes or driving is necessary) is a clear objective stressor. This claim is substantiated by the findings related to subjective stressors.

For commuter-train users, being satisfied with both the time it takes to reach the station and the time waiting at the station are negatively associated with stress. The unpredictability of the trip to the train station, and the tense time waiting for the train once there, may help explain these variables' importance. This is interesting, because for the two other transit mode types (subway and bus) satisfaction with the time to reach the station is not a concern. This may indicate that bus and subway stations are much more conveniently located; the trip to the station does not present a potentially stressful situation.

However, unlike train users, feeling that one's metro or bus trip is consistent is negatively associated with stress. Here, it indicates that for train users, once they are on the train the consistency of the trip is not an issue, but it is an issue for metro and bus users. For subway riders, comfort is also a concern. Having a high satisfaction with comfort is negatively associated with stress. Finally, like train users, for subway and bus users waiting times are significantly associated with stress. This indicates where perhaps the most stressful occurrence of any transit trip can occur, when waiting for the service, wondering when it is going to come.

5. Conclusion

The results of this study support previous research findings regarding commuting and stress (Manaugh & El-Geneidy, 2013). More importantly, this study demonstrates that the stress of commuting is intimately related to the mode being used: A general conception of commuting stress is not as powerful as mode-specific models. Similar to other studies, this study finds driving to be the most stressful commute. However, unlike previous studies, it finds that transit is not as stressful as driving.

Furthermore, our study confirms that commuting stress is caused by an interaction between objective stressors and mediators (time, control, and comfort) and subjective stressors which act as mediators (feelings, desires, and satisfaction). Driving is the most stressful mode because drivers must budget a considerable amount of extra time to deal with unexpected delays (their additional time budget has a mean of 21 min), and are more likely to be stressed when they are less satisfied with the time of their commute. This additional time budget indicates that they have, perhaps paradoxically, less control over their commute than commuters on other modes. Frequent and unpredictable occurrences require of them a peremptory stance toward their commute, where extra time becomes the best way to assure arriving to work or school on time.

This study also indicates that public transit users, although they are not as stressed as drivers, have a variety of stressors to contend with. The mode used to get to the station, satisfaction with the time to reach the station (for train riders), and waiting time at the station are all related to stress. All these factors can be seen as areas where control is wrested from the transit user. Unpredictability can occur at any moment in the transit experience, dissatisfaction with these moments may indicate a decrease in feelings of control over their situation, and thus more stress. Interestingly, only in the transit model is having used other modes related to less stress. It seems that having 'a plan B' for their commute lessens the stress of their trip. Transit agencies should consider policies that could better inform users (such as accurate bus or train arrival information) to lessen the stress experienced while traveling. Also, providing viable alternatives could further lessen transit stress.

Pedestrians are our least stressed mode group. For them, satisfaction with comfort and safety seem to have an important mediating effect on stress. These findings have notable policy implications. Active transportation modes are not only environmentally and socially more sustainable, they are also a less stressful way to travel. On way to increase pedestrian mode-share is to protect walkers from traffic and provide more pleasant and more comfortable streets to walk on. Furthermore, public transportation is also less stressful than driving, which is found to involve (somewhat perversely) less control for commuters. Increasing the predictability and range of transit options in an era of increasing driving unpredictability could lead to a greater transit mode share. Further investigation into the choice commuters have as a way to compare transportation modes should be studied in greater detail. For public transit users, measures of viable route options, including options not using the public transit network, can be gathered. This information could be easily provided to commuters, which would lessen the stress they experience and potentially increase their transit use. Having these choices, it is shown, has a negative effect on stress for transit users.

As a final note, a number of variables not included in this study impose limitations on our findings, and also offer avenues for future research. Our research uses one Likert scale response to measure stress. However, using a more standard stress scale while also incorporating outcomes of stress (missed days at work or school, mental or physical side effects) could make our findings more robust, and could be incorporated into future research. Also, a more direct question relating to a respondent's subjective experience of predictability might be useful. This variable could be used in conjunction with a respondent's additional time budget to measure both subjective and objective effects of predictability. In this study we collected data during a cold weather condition, so questions regarding good weather were excluded, as we felt some bias especially toward active modes. Future research collecting data to measure stress due to a commute need to be conducted across a longer time frame allowing for weather variation. Such expansion in the data collected period will make the reported levels of stress more accurate and representative of the impacts of different weather condition. Finally, subjective questions regarding a respondent's feelings of control or a personal estimate of how much effort (physical and mental) expended can greatly expand this study's findings.

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